

Linear Algebra — Exercises

Matrices

Exercise 1. For the following matrices A and B calculate the matrix $A \cdot B$:

(a) $A = \begin{pmatrix} 1 & 0 \\ 2 & 3 \\ 1 & 2 \\ -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & -2 & 2 \\ 1 & 5 & 3 \end{pmatrix}$ over the field $\mathbb{R}$

(b) $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 4 & 0 \\ 2 & 5 & 1 \\ 3 & 6 & 2 \end{pmatrix}$ over the field $\mathbb{Z}_7$

(c) $A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 3 & 2 & 3 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}$ over the field $\mathbb{Z}_5$

Exercise 2. Recall that a square matrix A is symmetric if and only if $A = A^T$. Is it true that multiplication of symmetric matrices is commutative?

Exercise 3. Show for matrices A and B (of appropriate sizes) that $(A \cdot B)^T = B^T \cdot A^T$.

Exercise 4. Use matrices to determine the dimension of the subspace W over the field $\mathbb{R}$:

(a) $W = \text{span}(\{(1, 2, 3), (1, -1, 2), (-1, 4, -1)\})$

(b) $W = \text{span}(\{(3, 1, 2, 1), (1, -1, 1, 2), (2, 3, 0, -3), (2, -1, 1, 2)\})$

Exercise 5. Calculate the following determinants:

(a) $\det \begin{pmatrix} 1 & 2 & -2 & 1 \\ -3 & 1 & 3 & 2 \\ -1 & 2 & 4 & 2 \\ 5 & -1 & 2 & 2 \end{pmatrix}$, where the matrix belongs to $\mathbb{R}_4^4$

(b) $\det \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 1 \\ 3 & 2 & 3 & 2 \\ 4 & 3 & 4 & 3 \end{pmatrix}$, where the matrix belongs to $(\mathbb{Z}_5)_4^4$

(c) $\det(A)$, where $A \in \mathbb{K}_5^5$ has the entries $a_{ij} = i - j$.

Exercise 6. The numbers 24026, 40262, 2624, 26240 and 62402 are divisible by 41. Prove that the following determinant is also divisible by 41:

$$\det \begin{pmatrix} 2 & 4 & 0 & 2 & 6 \\ 4 & 0 & 2 & 6 & 2 \\ 0 & 2 & 6 & 2 & 4 \\ 2 & 6 & 2 & 4 & 0 \\ 6 & 2 & 4 & 0 & 2 \end{pmatrix}.$$

Exercise 7. Let $y \in \mathbb{R}$, $n \in \mathbb{N}$ and let $A \in \mathbb{R}_n^n$ be defined in the following way: $a_{ij} = y$ for $j \neq n - i + 1$ and $a_{ij} = 0$ for $j = n - i + 1$. Find the determinant of A for $n = 3$, and then find the general formula for the determinant of A for an arbitrary $n \in \mathbb{N}$. Justify your answer.

Exercise 8. Show that for all matrices A and B of size 2×2 , $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Exercise 9. Show, for invertible matrices A and B , that $(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$.

Exercise 10. Calculate the inverse (if it exists) of each of the following matrices:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in (\mathbb{Z}_3)_4^4, \quad B = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 1 & 3 & 4 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 3 & 4 & 2 \end{pmatrix} \in \mathbb{R}_4^4, \quad C = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 1 & 3 & 4 & 1 \\ 0 & 3 & 1 & 0 \\ 1 & 3 & 4 & 2 \end{pmatrix} \in (\mathbb{Z}_5)_4^4.$$